SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:50**
Duration : 1 Hour

Application Based Question

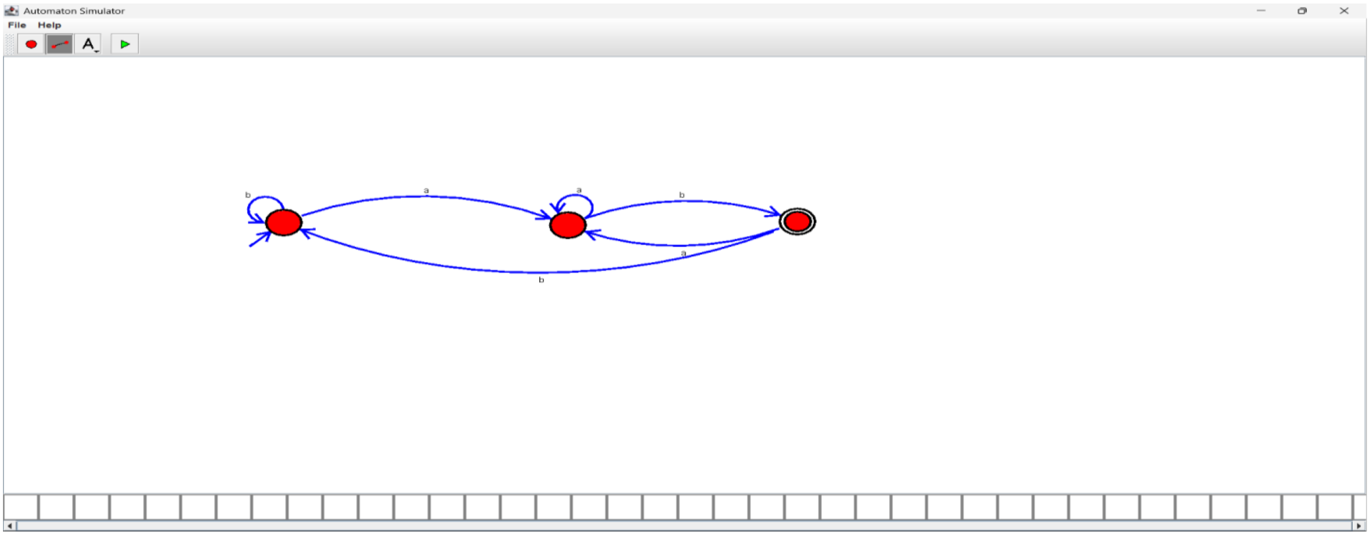
Code of Conduct:

IN Ramana (192372228) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

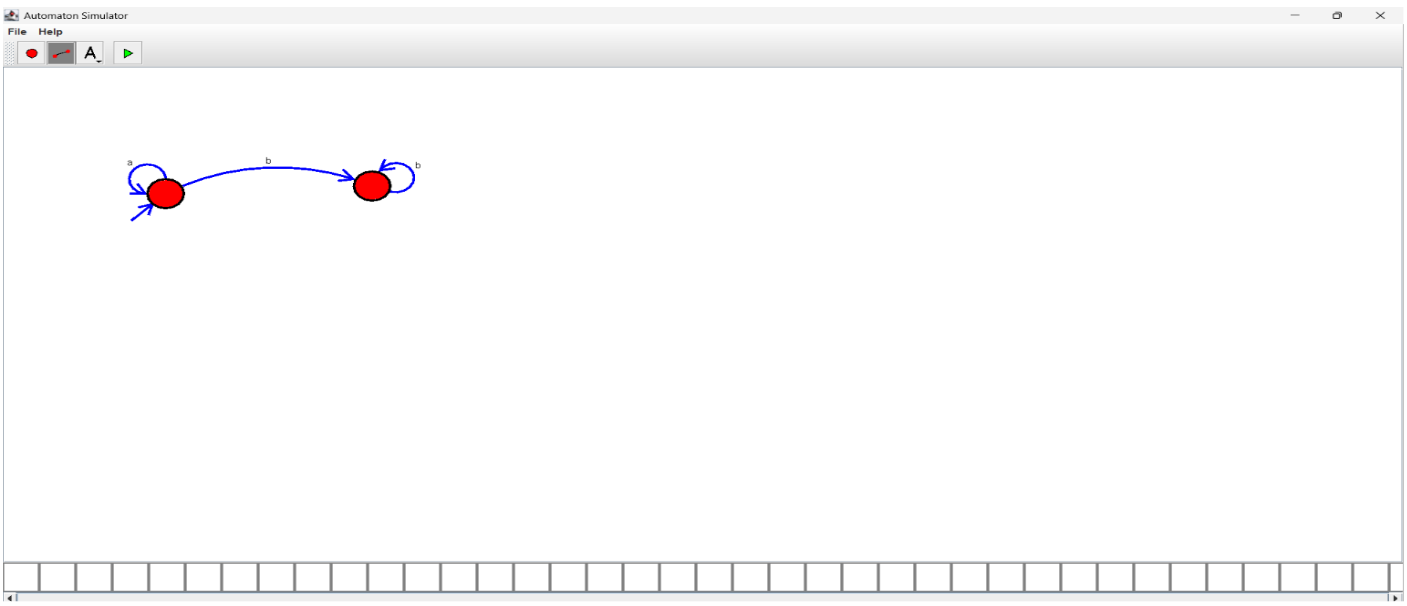
Signature of the student:

N Ramana

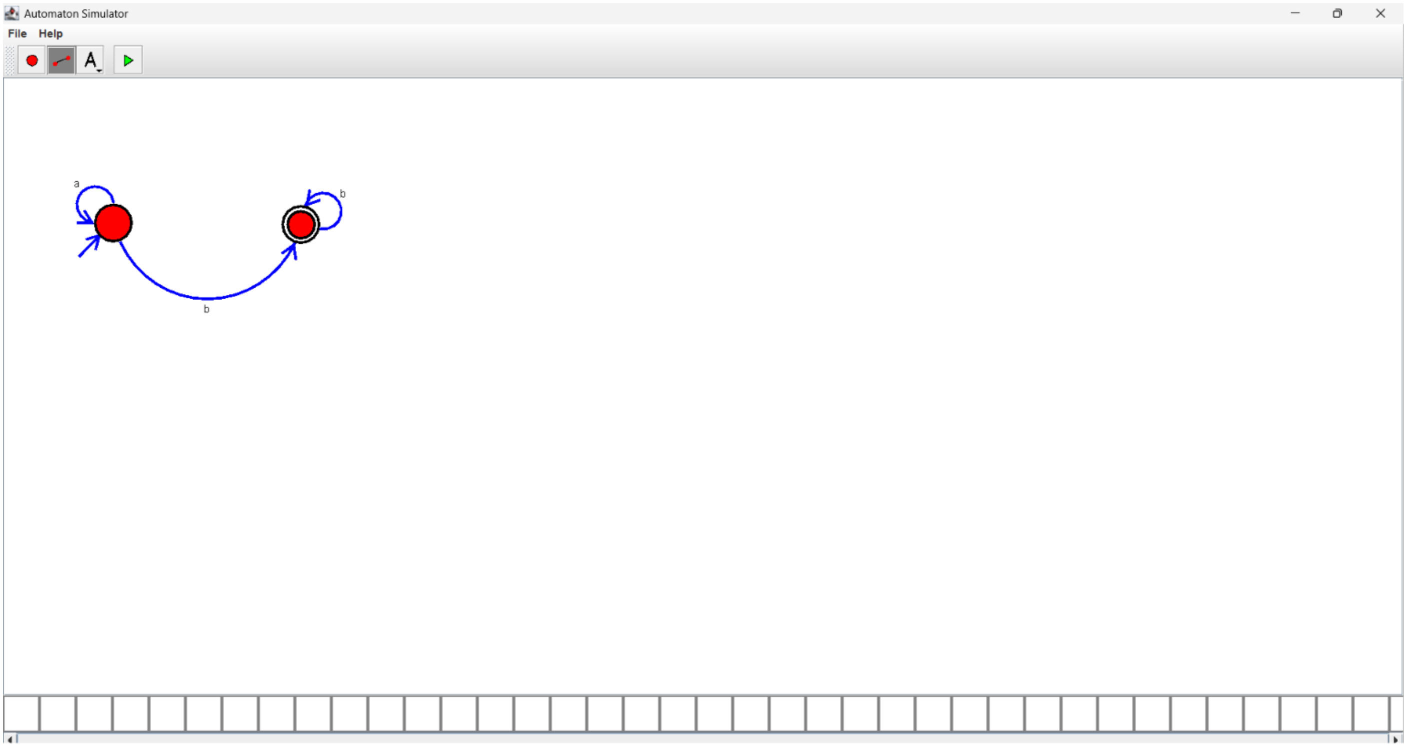
1.In modern digital systems such as embedded security devices, ATM machines, and authentication modules, password validation must be efficient and reliable.Design a **Deterministic Finite Automaton (DFA)** over the alphabet Σ = {0,1} to validate passwords that **end with the pattern "01"**.Further, **minimize the DFA** and analyze how the optimized design improves system efficiency in real-world hardware implementations



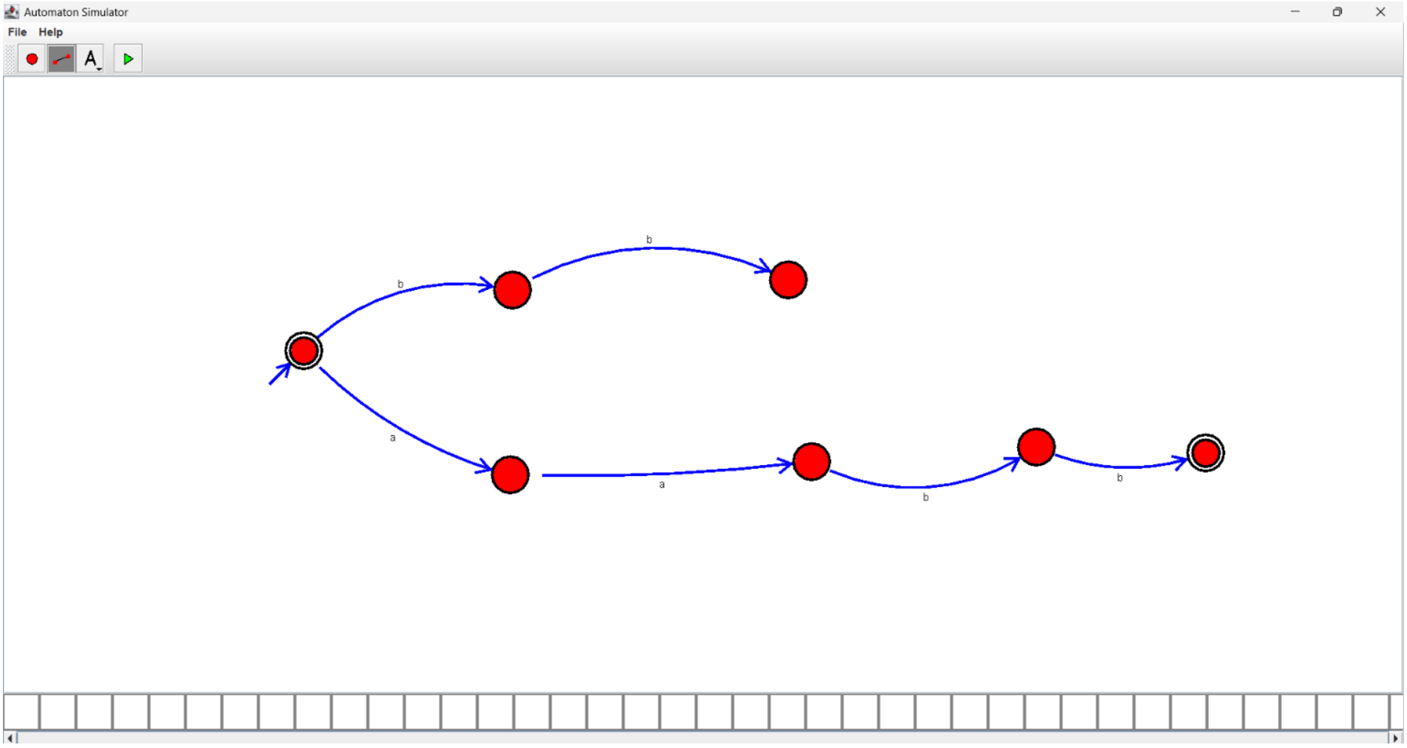
2.In web applications, validating user input is essential to ensure correctness, security, and performance.Design a regular expression to validate a specific input pattern "aaaabb" over the alphabet Σ = {a, b}.Further, analyze how regular languages and regular expressions are used to efficiently validate user inputs in real-world web applications**.**



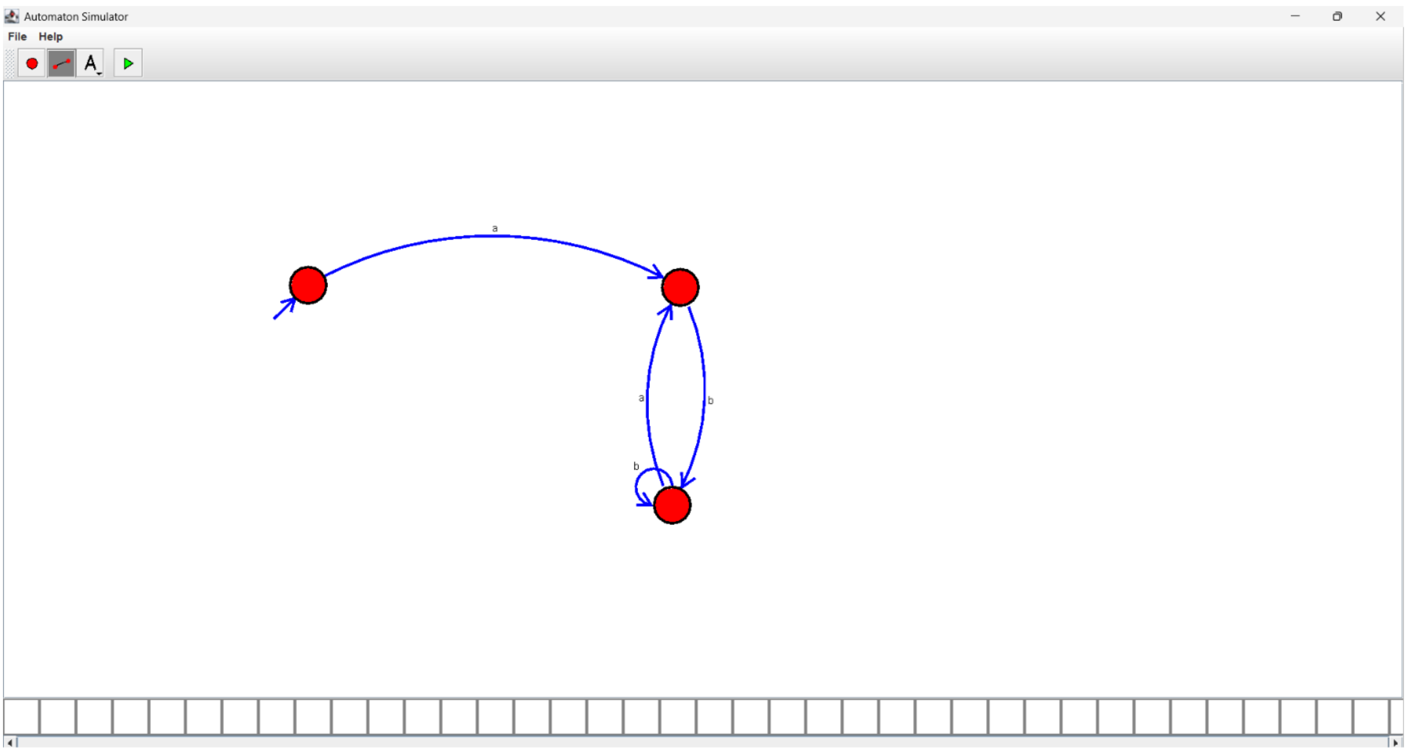
3. In modern computing systems such as intrusion detection systems, compilers, and real-time monitoring tools, pattern recognition plays a critical role.Demonstrate how the regular expression `a*b*` can be converted into an equivalent finite automaton (FA).Further, explain why regular expressions and finite automata are equivalent models and how they are effectively used in real-time detection systems.



4. In compiler design, not all patterns can be recognized using finite automata. Using the pumping lemma, demonstrate why the language L= { $a^n b^n$ | $n \geq 0$ } cannot be recognized by any finite automaton.Further, analyze the implications of this limitation in the design of modern compilers.



5.Modern systems such as intrusion detection, spam filtering, and input validation often require combining multiple pattern-matching rules.Explain how the **closure properties of regular languages** enable the combination of multiple detection rules.Further, justify why the resulting language after applying these operations **remains regular**, ensuring efficient implementation using finite automata



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RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	100		3	3	15
Accuracy of Answer	25	2.5			2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

2. Can enforce rules like:

- **Email format**
- **Password rules**
- **Input structure**